Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2026.XXXXXXX

LipScribe: Lip-Assisted Speech Recognition for Robust Live Captioning

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This work is submitted as the written project proposal for the six-week Capstone Project in Computer Vision and Deep Learning, Academic Year Advent 2026, Group 4.

ABSTRACT Automatic Speech Recognition systems fail when background noise corrupts the audio signal. Lip movements remain completely unaffected by acoustic noise and therefore provide a reliable complementary channel. LipScribe is a multimodal system that improves transcription by combining audio understanding with lip reading. The system crops the mouth region using MediaPipe Face Mesh, extracts spatiotemporal features with a 3D Convolutional Neural Network, processes audio with recurrent networks, and fuses both streams using attention that adapts to noise level. Training uses the GRID corpus with controlled noise injection. Evaluation is performed with Word Error Rate and Character Error Rate across multiple signal-to-noise ratios. The primary beneficiaries are deaf and hard-of-hearing users who depend on accurate live captions in lectures, meetings and public spaces.

INDEX TERMS Audiovisual speech recognition, lip reading, multimodal fusion, connectionist temporal classification, MediaPipe, live captioning, assistive technology, noise robustness

TABLE OF ABBREVIATIONS

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3D-CNN	Three-Dimensional Convolutional Neural Network 2
ASR	Automatic Speech Recognition 1
BiLSTM	Bidirectional Long Short-Term Memory 2
CER	Character Error Rate 2
CTC	Connectionist Temporal Classification 2
MFCC	Mel-Frequency Cepstral Coefficients 2
ROI	Region of Interest 2
SNR	Signal-to-Noise Ratio 2
WER	Word Error Rate 2

I. INTRODUCTION

A. BACKGROUND

Traditional Automatic Speech Recognition (ASR) systems work well in quiet rooms but degrade sharply when background noise, overlapping speakers or muffled speech corrupt the audio signal.[1] Lip movements, however, are completely unaffected by acoustic noise. By watching the mouth, a computer can still obtain useful information about what is being said. Combining the audio signal with visual lip information therefore offers a more reliable route to accurate live captions, especially in real-world noisy environments.

B. PROBLEM STATEMENT

Most current live-captioning tools rely only on audio. In noisy classrooms, public transport, industrial sites or outdoor events these tools produce high error rates. The central problem addressed by this project is:

> *How can we improve transcription by combining audio understanding with lip reading so that captions remain usable even when the audio channel is severely degraded?*

[1]Automatic Speech Recognition (ASR) means converting spoken words into written text.

C. RESEARCH OBJECTIVES

The project has four concrete objectives:

1) **Disambiguate homophenes**: Learn to distinguish lip shapes that look almost identical but correspond to different letters or words (for example "mat" versus "bat").[2]
2) **Model temporal word sequences**: Process continuous sequences of video frames so that the system recognises whole words and short phrases rather than isolated letters.
3) **Detect emotion from speaking style**: Capture emotional information from the dynamics of lip and facial movement while the person is speaking.
4) **Identify the spoken language**: Recognise which language the speaker is using by jointly analysing lip movements and audio.

D. JUSTIFICATION

More than 466 million people worldwide live with hearing loss and depend on captions or lip reading for everyday communication. Human sign-language interpreters are scarce and commercial assistive devices remain expensive. A practical system that improves live captions in noisy conditions can increase accessibility in education, healthcare and public services. From a technical perspective the project is well suited to a six-week computer-vision capstone: it requires real data handling, a clear baseline, transfer learning, and the integration of at least two advanced capabilities (spatiotemporal visual encoding and adaptive multimodal fusion).

II. LITERATURE REVIEW

Combining audio and visual information has repeatedly been shown to improve speech recognition under noise. When the acoustic signal is corrupted, visual lip features remain stable and the model can shift attention toward the mouth region [1], [2].

Accurate isolation of the mouth is a critical first step. MediaPipe Face Mesh detects 468 three-dimensional facial landmarks in real time and enables precise cropping of the lip Region of Interest (ROI), removing background clutter such as the nose, cheeks and facial hair [3].[3]

Reading a sequence of frames is far more powerful than examining single images. Three-Dimensional Convolutional Neural Network (3D-CNN) networks capture both the shape of the lips and how they move over time. These features are commonly passed to Bidirectional Long Short-Term Memory (BiLSTM) layers so that the model can learn the full temporal pattern of a word [4], [5].

Connectionist Temporal Classification (CTC) loss is widely adopted because it removes the need for expensive frame-level alignments. CTC allows a network to map a variable-length sequence of visual and audio features directly onto a sequence of characters [6], [7].[4]

A persistent difficulty is the existence of homophenes – different sounds that produce nearly identical lip shapes (for example /p/ versus /b/). Single-frame visual models cannot resolve this ambiguity; temporal context and cross-modal attention are required [1].

III. METHODOLOGY

A. DATASET AND NOISE PREPARATION

The primary dataset is the GRID audiovisual sentence corpus, which contains 34 speakers and approximately 33 000 short sentences with clean frontal video and aligned audio. Synthetic noise from the MUSAN corpus will be mixed into the audio at controlled Signal-to-Noise Ratio (SNR) levels (clean, 10 dB, 5 dB, 0 dB and -5 dB).[5]

B. MOUTH CROPPING

MediaPipe Face Mesh will be applied to every video frame. The lip keypoints are used to compute a tight bounding box, which is then cropped and resized to a fixed resolution (96×96 or 112×112).

C. DUAL-STREAM ARCHITECTURE

- **Visual stream**: Cropped mouth sequences pass through a 3D convolutional front-end followed by a bidirectional LSTM.
- **Audio stream**: Raw audio is converted to log-Mel spectrograms and Mel-Frequency Cepstral Coefficients (MFCC) features, then encoded by a bidirectional LSTM.

D. MULTIMODAL FUSION

Audio and visual features are combined by an attention-based fusion module that automatically gives more weight to the lips when the audio is noisy. During training we also use modality dropout so the network cannot ignore the visual stream.

E. BASELINES AND ADVANCED TECHNIQUES

We first train three simple baselines (audio-only, visual-only, and simple concatenation). The final system adds two advanced capabilities: (1) a 3D convolutional visual encoder and (2) adaptive cross-modal attention with modality dropout. Sequence prediction uses CTC loss with beam search.

F. EVALUATION

Performance is measured by Word Error Rate (WER) and Character Error Rate (CER) on held-out speakers at every

[2]Homophenes are different sounds that produce almost the same lip shape.
[3]Region of Interest (ROI) is the small rectangular area of the image that contains only the mouth.
[4]Connectionist Temporal Classification (CTC) is a special training method that lets the model learn the alignment between video frames and text automatically.
[5]Signal-to-Noise Ratio (SNR) tells us how strong the voice is compared with the background noise. Lower SNR means more noise.

tested SNR. Ablation studies will show the contribution of each advanced component.

IV. EXPECTED RESULTS

Under clean conditions the fused model is expected to achieve a Word Error Rate below 10 %. As noise increases, the visual stream should increasingly compensate for the degraded audio, producing a clear advantage over the audio-only baseline at 0 dB and −5 dB SNR. The final deliverable will include a clean inference script and a simple web demonstration.

ACKNOWLEDGMENT

The authors thank the course lecturers for guidance on problem formulation, evaluation standards and responsible computer-vision practice.

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